Software Requirements Specification (SRS)

Project: Smart Parking Management System (SPMS)
Version: 1.1
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Revision history

Version	Date	Author	Change summary	Approval
1.0	07-10-2026	Team 06	Initial SRS with use-case diagrams embedded	coordinator@pes.edu
1.1	08-11-2026	Team 06	As-built update after Sprint 2: RTM status filled from the test cycle; deviations listed in docs/Deviations.md and section 9	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator	Coordinator	coordinator@pes.edu	11/10/2026

Table of Contents

1. Introduction
2. Overall description
3. External interfaces
4. System features (detailed)
5. Non-functional requirements (detailed)
6. Quality attributes & Acceptance tests
7. UML Use-Case Diagram

8. Requirements Traceability Matrix (RTM)

1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for the Smart Parking Management System (SPMS). It defines functional and non-functional requirements, interfaces, security requirements and verification criteria for the development team, testers and course evaluators.

1.2 Scope

Covers tracking of parking availability, vehicle entry and exit, advance slot reservations, automatic allocation of suitable parking spaces, fee calculation and payments (online in gateway test mode and cash), receipts, notifications and occupancy/revenue information for administrators. Excludes physical hardware such as boom barriers, ANPR cameras and slot sensors (their events are entered through the attendant console), real-money settlement and native mobile apps.

1.3 Audience

Developers (Team 06), QA/Test Engineers, Parking Administrators, Course Instructor and Assessment Evaluators.

1.4 Definitions

List of acronyms: SPMS, UI, API, REST, JWT, RBAC, QR, UPI, TLS, WSS, ORM, CSV, FR, NFR, RTM, RPO, RTO. Lot = parking facility; Level = floor; Slot = single parking space; Session = one stay from entry to exit; Grace period = free initial minutes; No-show = reservation not used within 15 min of its start.

2. Overall description

2.1 Product perspective

SPMS is a new web-based system used by drivers, parking attendants and administrators. It includes a React web front end (driver portal, attendant gate console, admin dashboard), a Python FastAPI backend, a PostgreSQL database, Redis for caching and live updates, and integrations with a payment gateway (test mode) and an email server.

2.2 Major product functions (detailed)

- User registration, login and role-based access
- Parking lot, level, slot and tariff management
- Real-time slot availability and occupancy display
- Advance slot reservation with double-booking prevention
- Automatic allocation of the most suitable slot
- Vehicle entry and exit recording (plate number / reservation QR)
- Fee calculation, online / cash payment and digital receipts
- Occupancy and revenue reports, CSV export and email notifications

2.3 User roles and characteristics (expanded)

- Driver: basic smartphone/internet skills; wants to find, reserve and pay for a slot quickly.
- Parking Attendant: works at the gate under time pressure; needs fast entry/exit screens and clear errors.
- Administrator: manages lots, slots and tariffs; needs dashboards, reports and audit trails.

2.4 Operating environment

Linux server (Docker) running Python 3.11/FastAPI, PostgreSQL 15 and Redis 7; clients use the latest Chrome, Edge, Firefox or Safari on desktop, tablet or mobile over HTTPS.

2.5 Constraints

Python backend and React frontend as agreed by the team, payment gateway in sandbox/test mode only, personal-data protection (DPDP Act 2023 principles), open-source tools only, completion within the semester by a team of four.

3. External interface requirements

3.1 User interfaces

Responsive web UI: driver portal (availability, reservation, payment, history), tablet-friendly attendant gate console (Entry/Exit, plate input, QR scan, fee due) and admin dashboard (CRUD screens and charts). Accessibility: keyboard navigation, screen-reader labels and WCAG 2.1 AA colour contrast.

3.2 Hardware interfaces

- Standard PC / tablet with a browser at the gate
- Optional USB/Bluetooth QR scanner (keyboard / HID mode)
- Boom barrier, ANPR camera and slot sensors: out of scope (simulated via Gate API)

3.3 Software interfaces

- PostgreSQL 15 database (via SQLAlchemy ORM) for all persistent data.
- Redis 7 for availability cache, locks, pub/sub and Celery task queue.
- Payment Gateway REST API (Razorpay test mode) for orders, refunds and signed webhooks; SMTP server for emails.

3.4 Communications

- HTTPS (TLS 1.2+) REST/JSON API (/api/v1) and WebSocket Secure (WSS) for live availability updates.
- Signed (HMAC-SHA256) webhooks from the payment gateway; SMTP over TLS for email; retry with backoff for external calls.

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow SPMS-F-###.

4.1 User Registration & Authentication

Description: Allow drivers to create accounts and all users (Driver, Parking Attendant, Administrator) to sign in securely with role-based access.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
SPMS-F-001	The system shall allow a driver to register with name, email, mobile number and	Functional	High	Driver	Valid form creates an account and returns 201; duplicate email returns 409 with	Email service for welcome mail

	password, rejecting duplicate email/mobile.				a clear message. Test: TC-AUTH-01	
SPMS-F-002	The system shall authenticate users with email and password and issue a signed JWT access token (30 min) and a refresh token.	Functional	High	Security	Valid credentials return tokens; invalid credentials return 401 with a generic message. Test: TC-AUTH-02	Depends on SPMS-SR-002/003
SPMS-F-003	The system shall enforce role-based access control (Driver, Attendant, Admin) on every API endpoint and UI screen.	Functional	High	Security / Admin	A Driver token calling an Admin endpoint receives 403; menu items hidden by role. Test: TC-AUTH-03	Applies to all modules
SPMS-F-004	The system shall lock an account for 15	Functional	High	Security	6th attempt within lock window is	Audit Logger

	minutes after 5 consecutive failed login attempts and record an audit event.				refused; audit entry exists. Test: TC-AUTH-04	

4.2 Parking Lot & Slot Management

Description: Let the administrator model the physical parking facility (lots → levels → slots) and its pricing.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
SPMS-F-010	The system shall allow the administrator to create, update and deactivate parking lots, levels and slots, each slot having a type (2-wheeler, 4-wheeler, EV-charging, accessible).	Functional	High	Parking owner	CRUD operations persist and appear in availability within 5 s. Test: TC-LOT-01	—
SPMS-F-011	The system	Functional	High	Parking owner	New tariff is applied	Used by Billing

	shall allow the administrator to configure tariffs per lot and vehicle type (grace period, hourly rate, daily cap) with an effective date.				only to sessions starting after its effective time. Test: TC-LOT-02	
SPMS-F-012	The system shall allow an administrator or attendant to mark a slot 'Under Maintenance', excluding it from allocation and availability counts.	Functional	Medium	Operations	Slot in maintenance is never allocated; free count decreases by one. Test: TC-LOT-03	Allocation Engine

4.3 Real-time Availability

Description: Provide accurate, live occupancy information to drivers and staff.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
SPMS-F-	The	Function	High	Driver	Displaye	—

020	system shall display the number of free slots per lot, level and slot type to all users, including unauthenticated visitors.	al			d counts equal database state for a seeded lot. Test: TC-AVL-01	
SPMS-F-021	The system shall push slot status changes to connected clients within 5 seconds of an entry, exit, reservation or maintenance event.	Functional	High	Driver / Operations	Change visible on a second browser within 5 s. Test: TC-AVL-02	WebSocket + Redis pub/sub

4.4 Slot Reservation

Description: Let drivers book a slot in advance and manage their bookings.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
SPMS-F-030	The system shall allow a	Functional	High	Driver	Valid request returns 201 with	Allocation Engine, Payment

	driver to reserve a slot for a registered vehicle in a chosen lot for a future time window (30 min – 24 h long, up to 7 days ahead).				slot code and QR token after advance payment. Test: TC-RES-01	
SPMS-F-031	The system shall prevent double booking: no two active reservations or sessions may overlap in time on the same slot.	Functional	High	Business	Of 50 concurrent requests for the last free slot exactly one succeeds. Test: TC-RES-02	DB exclusion constraint
SPMS-F-032	The system shall allow a driver to modify or cancel a reservation before its start time, refunding 100% of	Functional	Medium	Driver / Business	Refund amount follows policy; slot released immediately. Test: TC-RES-03	Payment gateway refund API

	the advance if cancelled ≥ 1 h before start, otherwise 50%.					
SPMS-F-033	The system shall auto-cancel a reservation and release the slot if the vehicle has not entered within 15 minutes after the reservation start time (no-show).	Functional	Medium	Operations	Reservation status becomes NO_SHOW and slot is free at start + 15 min. Test: TC-RES-04	Scheduled job (Celery beat)

4.5 Suitable Slot Allocation

Description: Automatically assign the best available slot for each vehicle.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
SPMS-F-040	The system shall automatically allocate a free slot matching	Functional	High	Driver / Operations	For seeded data the allocated slot equals the expected nearest	—

	the vehicle type and accessibility need, preferring the slot with the least distance to the entrance.				compatible slot. Test: TC-ALC-01	
SPMS-F-041	The system shall allow an attendant to override the automatic allocation with another free compatible slot, recording a reason in the audit log.	Functional	Medium	Operations	Override succeeds only for free compatible slots; audit entry stored. Test: TC-ALC-02	Audit Logger

4.6 Vehicle Entry & Exit

Description: Record vehicles entering and leaving the facility at the gate.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
SPMS-F-050	The system shall allow an attendant to record	Functional	High	Operations	Entry creates an OPEN parking session and	Allocation Engine

	vehicle entry by plate number (and optional reservation QR), storing the timestamp and issuing a ticket ID with the allocated slot code.				marks the slot OCCUPIED. Test: TC-GATE-01	
SPMS-F-051	The system shall validate a reservation QR at entry: correct lot, within the reservation window (±15 min) and status CONFIRMED.	Functional	High	Business	Expired / wrong-lot / cancelled QR is rejected with reason. Test: TC-GATE-02	Reservation Service
SPMS-F-052	The system shall reject entry of a plate number that already	Functional	High	Business	Second entry for same plate returns 409 DUPLICATE_ENTRY.	—

	has an open parking session.				Test: TC-GATE-03	
SPMS-F-053	The system shall record vehicle exit, compute parking duration and fee, close the session and free the slot once payment is settled.	Functional	High	Operations	Exit closes session; slot becomes FREE; availability updated. Test: TC-GATE-04	Billing & Payment

4.7 Billing & Payments

Description: Calculate parking charges and collect payment online or at the counter.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
SPMS-F-060	The system shall compute the fee as 0 within the grace period, otherwise ceil(hours) × hourly rate, capped	Functional	High	Business	Fee matches expected values for boundary cases (grace, 1 h + 1 min, > 24 h). Test: TC-PAY-01	Tariff configuration

	at the daily cap per 24 h, minus any advance already paid.					
SPMS-F-061	The system shall accept online payment (UPI / card via the payment gateway's hosted checkout) and cash payments recorded by the attendant.	Functional	High	Driver / Operations	Successful payment marks session/reservation PAID. Test: TC-PAY-02	Payment gateway (test mode)
SPMS-F-062	The system shall generate a digital receipt (receipt no., plate, entry/exit time, duration, amount, method) viewable in the app and	Functional	Medium	Driver	Receipt fields match session and payment data. Test: TC-PAY-03	—

	downloadable as PDF.					
SPMS-F-063	The system shall not close a session with an unpaid fee unless an administrator records a waiver with a reason.	Functional	Medium	Business	Exit without payment is blocked (402); waiver is audited. Test: TC-PAY-04	Audit Logger

4.8 Reports, History & Notifications

Description: Give administrators occupancy/revenue insight and keep drivers informed.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
SPMS-F-070	The system shall show administrators a dashboard of occupancy %, revenue, average stay and peak hours per lot for a selected date	Functional	Medium	Parking owner	Dashboard figures match SQL computed from test data. Test: TC-RPT-01	—

	range.					
SPMS-F-071	The system shall allow administrators to export occupancy and revenue reports as CSV.	Functional	Low	Parking owner	CSV opens in a spreadsheet and totals match the dashboard. Test: TC-RPT-02	—
SPMS-F-072	The system shall email the driver on reservation confirmation, cancellation/refund, and 15 minutes before the reservation starts.	Functional	Medium	Driver	Emails captured in test SMTP inbox with correct content. Test: TC-NOT-01	SMTP server
SPMS-F-073	The system shall allow a driver to view their past reservations, parking sessions and receipts.	Functional	Medium	Driver	History lists only the logged-in driver's records. Test: TC-HIS-01	—

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs SPMS-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
SPMS-NF-001	95% of API requests shall complete within 2 seconds with 200 concurrent users under normal load.	Performance	High	Locust load test: p95 ≤ 2 s. Test: TC-PERF-01
SPMS-NF-002	A gate entry or exit transaction (request to ticket/receipt) shall complete within 3 seconds.	Performance	High	Measured over 100 gate operations: max ≤ 3 s. Test: TC-PERF-02
SPMS-NF-003	The system shall be available 99% of operating hours (06:00–23:00); the database shall be backed up daily (RPO ≤ 24 h, RTO ≤ 2 h).	Reliability / Availability	High	Uptime monitor report; restore drill succeeds within 2 h. Test: TC-REL-01
SPMS-NF-004	A first-time driver shall be able to complete a reservation in at most 4 screens / 2	Usability / Accessibility	Medium	Usability test with 5 users; Lighthouse accessibility ≥ 90. Test: TC-UX-01

	minutes; UI shall be responsive from 360 px to 1920 px wide and meet WCAG 2.1 AA colour contrast.			
SPMS-NF-005	The system shall support at least 10 lots, 5,000 slots and 1,000 reservations per day without schema changes, with a stateless API that can be scaled horizontally.	Scalability	Medium	Load test with seeded volume meets NF-001. Test: TC-PERF-03
SPMS-NF-006	Backend code shall follow PEP 8 (flake8 clean), have ≥ 70% unit-test line coverage and auto-generated OpenAPI documentation.	Maintainability	Medium	CI pipeline report shows coverage ≥ 70% and 0 lint errors. Test: TC-MNT-01

5.1. Security

5.1.1 Security Objectives

- SO-1 Confidentiality – Protect users' personal data (name, email, phone, vehicle plate numbers) and payment information from unauthorised disclosure.
- SO-2 Integrity – Ensure slot states, parking sessions, fees and payment confirmations cannot be tampered with (e.g., fee manipulation or forged payment success).
- SO-3 Accountability – Make every privileged action (allocation override, fee waiver, tariff change) traceable to an authenticated user.
- SO-4 Availability – Resist brute-force and request-flooding abuse so that gate operations continue during peak hours.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
SPMS-SR-001	All network traffic (browser ↔ API, API ↔ payment gateway/SMTP) shall use TLS 1.2 or higher, with HSTS enabled.	Security	High	SSL Labs / testssl scan shows no TLS < 1.2. Test: TC-SEC-01
SPMS-SR-002	Passwords shall be stored only as bcrypt hashes (cost ≥ 12) and shall never be logged or returned by any API.	Security	High	DB inspection and log review show no plaintext. Test: TC-SEC-02
SPMS-SR-003	Access tokens shall be signed JWTs expiring in 30 minutes; refresh tokens shall be rotated on use and revoked on logout.	Security	High	Expired / tampered / revoked token returns 401. Test: TC-SEC-03
SPMS-SR-004	All inputs shall be validated server-side with Pydantic schemas and all database access shall use parameterised ORM queries; output shall be encoded to prevent XSS.	Security	High	OWASP ZAP scan: 0 high findings for SQLi/XSS. Test: TC-SEC-04
SPMS-SR-005	Card data shall never be handled or stored by SPMS (gateway hosted checkout); a payment shall be marked successful only after HMAC-SHA256 webhook signature	Security	High	Forged webhook is rejected (400) and payment stays PENDING. Test:

	verification.			TC-SEC-05
SPMS-SR-006	Login, registration and payment endpoints shall be rate-limited (e.g., 10 requests/min per IP for login).	Security	Medium	11th request in a minute returns 429. Test: TC-SEC-06
SPMS-SR-007	Privileged actions shall be written to an append-only audit log containing user, role, timestamp, action and before/after values, retained for 1 year.	Security	Medium	Override/ waiver/tariff change creates audit entry; log rows cannot be updated via API. Test: TC-SEC-07

6. Quality attributes & Acceptance tests

- Exit criteria for acceptance: All high-priority functional and security requirements implemented and verified, no critical NFR failures, and RTM shows all test cases passed.
- Acceptance test suites: Authentication, Lot & Slot Management, Availability, Reservation, Allocation, Gate Entry/Exit, Payments, Reports & Notifications, Performance, Reliability, Usability, Maintainability and Security tests.

7. System models and diagrams

7.1 UML Use-Case diagram

Two UML use case diagrams are provided: (1) Driver services and (2) Gate operations & administration.

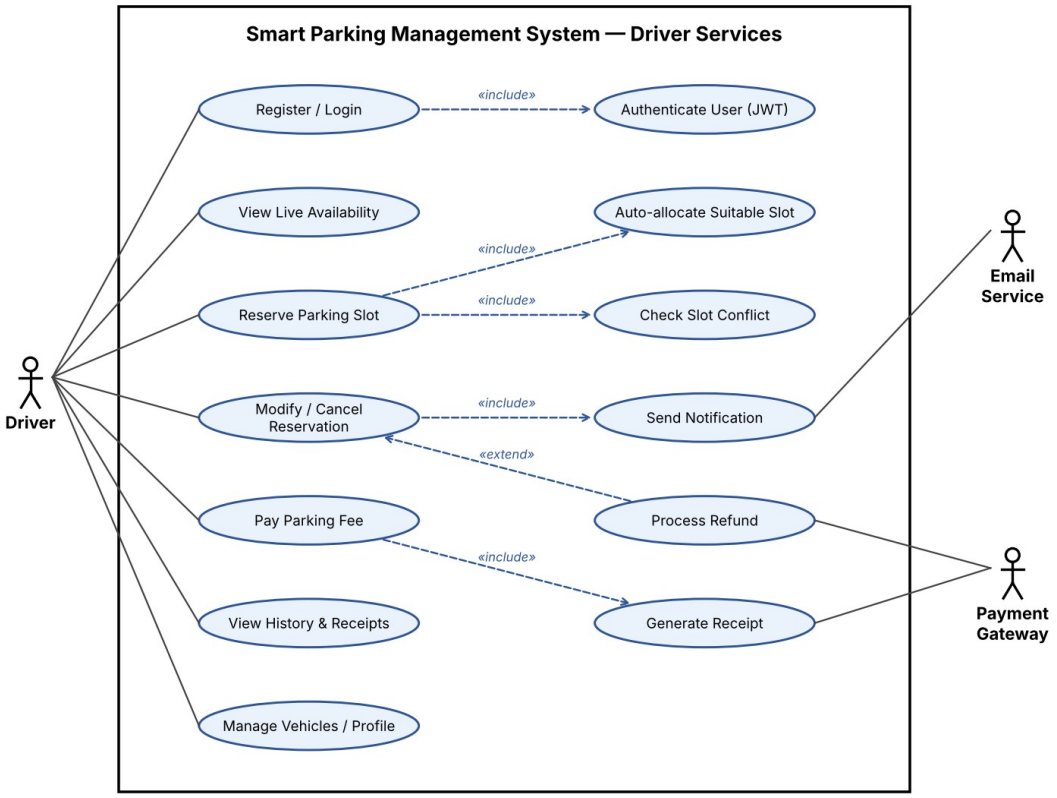


Figure 1: Use-case diagram – Driver services

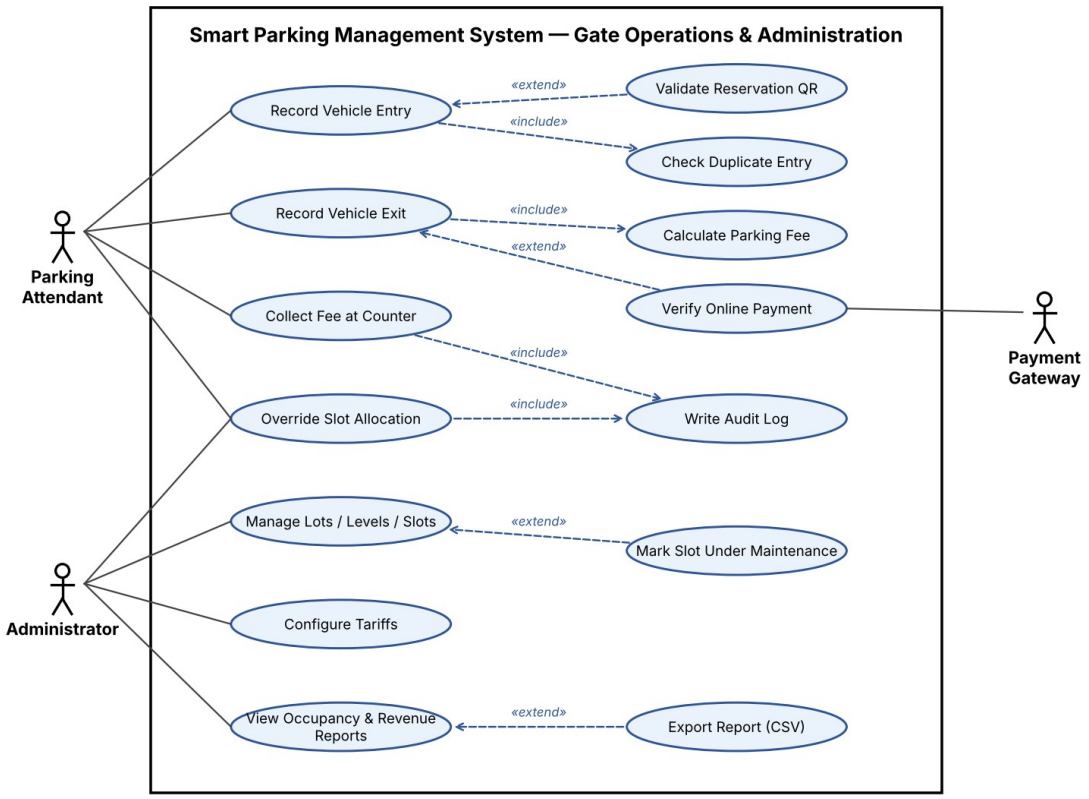


Figure 2: Use-case diagram – Gate operations & administration

8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
SPMS-F-001	Driver registration	4.1 / DS-AUTH-01	Auth & User Service	TC-AUTH-01	P	
SPMS-F-002	Login & JWT issue	4.1 / DS-AUTH-01	Auth & User Service	TC-AUTH-02	P	
SPMS-F-003	Role-based access control	4.1 / DS-AUTH-02	API Router / Auth MW	TC-AUTH-03	P	
SPMS-F-004	Account lockout after 5 failures	4.1 / DS-AUTH-01	Auth & User Service	TC-AUTH-04	P	
SPMS-F-010	Manage lots / levels / slots	4.2 / DS-LOT-01	Lot & Slot Management	TC-LOT-01	P	
SPMS-F-011	Configure tariffs	4.2 / DS-LOT-02	Lot & Slot Management	TC-LOT-02	P	
SPMS-F-012	Slot maintenance status	4.2 / DS-LOT-01	Lot & Slot Management	TC-LOT-03	P	
SPMS-F-020	Display free-slot counts	4.3 / DS-AVL-01	Availability Service	TC-AVL-01	P	
SPMS-F-021	Live status push ≤ 5 s	4.3 / DS-AVL-01	Availability Service	TC-AVL-02	P	
SPMS-F-030	Reserve a slot	4.4 / DS-RES-01	Reservation Service	TC-RES-01	P	
SPMS-F-	Prevent	4.4 / DS-	Reservati	TC-RES-	P	

031	double booking	RES-02	on Service	02		
SPMS-F-032	Modify / cancel with refund	4.4 / DS-RES-01	Reservation Service	TC-RES-03	P	
SPMS-F-033	Auto-release no-shows	4.4 / DS-RES-03	Reservation Service	TC-RES-04	P	
SPMS-F-040	Auto-allocate suitable slot	4.5 / DS-ALC-01	Slot Allocation Engine	TC-ALC-01	P	
SPMS-F-041	Attendant allocation override	4.5 / DS-ALC-01	Slot Allocation Engine	TC-ALC-02	P	
SPMS-F-050	Record vehicle entry	4.6 / DS-GATE-01	Entry/ Exit Gate Service	TC-GATE-01	P	
SPMS-F-051	Validate reservation QR	4.6 / DS-GATE-01	Entry/ Exit Gate Service	TC-GATE-02	P	
SPMS-F-052	Reject duplicate entry	4.6 / DS-GATE-01	Entry/ Exit Gate Service	TC-GATE-03	P	
SPMS-F-053	Record vehicle exit	4.6 / DS-GATE-02	Entry/ Exit Gate Service	TC-GATE-04	P	
SPMS-F-060	Fee calculation	4.7 / DS-PAY-01	Billing & Payment Service	TC-PAY-01	P	
SPMS-F-061	Online & cash payment	4.7 / DS-PAY-02	Billing & Payment Service	TC-PAY-02	P	
SPMS-F-062	Digital receipt	4.7 / DS-PAY-01	Billing & Payment Service	TC-PAY-03	P	
SPMS-F-063	Block unpaid exit / waiver	4.7 / DS-PAY-01	Billing & Payment Service	TC-PAY-04	P	
SPMS-F-070	Occupancy &	4.8 / DS-RPT-01	Reporting &	TC-RPT-01	P	

	revenue dashboard		Analytics			
SPMS-F-071	CSV export	4.8 / DS-RPT-01	Reporting & Analytics	TC-RPT-02	P	
SPMS-F-072	Email notifications	4.8 / DS-NOT-01	Notification Service	TC-NOT-01	P	
SPMS-F-073	Driver history & receipts	4.8 / DS-RPT-02	Reporting & Analytics	TC-HIS-01	P	
SPMS-NF-001	API response time	5 / DS-PERF-01	FastAPI / PostgreSQL	TC-PERF-01	P	
SPMS-NF-002	Gate transaction time	5 / DS-PERF-01	Gate Service	TC-PERF-02	P	
SPMS-NF-003	Availability & backup	5 / DS-OPS-01	Deployment, backups	TC-REL-01	P	
SPMS-NF-004	Usability & accessibility	5 / DS-UX-01	React UI	TC-UX-01	P	
SPMS-NF-005	Scalability	5 / DS-PERF-01	Architecture	TC-PERF-03	P	
SPMS-NF-006	Maintainability	5 / DS-OPS-02	CI (GitHub Actions)	TC-MNT-01	P	
SPMS-SR-001	TLS 1.2+ everywhere	5.1.2 / DS-SEC-01	Cross-cutting	TC-SEC-01	A	SO-1, SO-2; TLS at the reverse proxy in deployment (deviation D-08)
SPMS-SR-002	bcrypt password hashing	5.1.2 / DS-SEC-01	Cross-cutting	TC-SEC-02	P	SO-1
SPMS-SR-003	JWT expiry &	5.1.2 / DS-SEC-	Cross-cutting	TC-SEC-03	P	SO-1, SO-3

	revocation	01				
SPMS-SR-004	Input validation / no SQLi-XSS	5.1.2 / DS-SEC-01	Cross-cutting	TC-SEC-04	P	SO-2
SPMS-SR-005	No card data; signed webhooks	5.1.2 / DS-SEC-01	Cross-cutting	TC-SEC-05	P	SO-1, SO-2
SPMS-SR-006	Rate limiting	5.1.2 / DS-SEC-01	Cross-cutting	TC-SEC-06	P	SO-4
SPMS-SR-007	Append-only audit log	5.1.2 / DS-SEC-01	Cross-cutting	TC-SEC-07	P	SO-3

9. Changes in version 1.1

No requirement was removed or weakened. RTM status legend: N = not yet tested, P = passed in the test cycle of 07-08 Nov 2026, A = accepted with a documented deviation. Implementation choices that differ from the SAD v1.0 (plain HTML/JS pages instead of React, SQLite instead of PostgreSQL, in-process jobs instead of Redis/Celery, mock payment gateway, email outbox) are listed in docs/Deviations.md. SPMS-SR-001 (TLS) is met at deployment; the local demo runs on http://localhost.